

HW#3; Problems Statement:

- 3-48** For the circuit in Figure P3-48 find its Thévenin and Norton equivalent circuits.

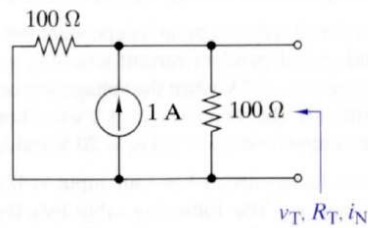


FIGURE P3-48

- 3-49(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-49.
(b) Use the equivalent circuit found in (a) to find i_L if $R_L = 22 \text{ k}\Omega$.

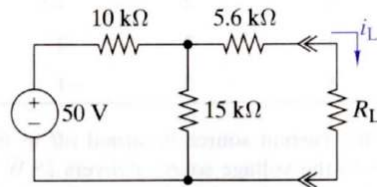


FIGURE P3-49

- 3-58** Assume that Figure P3-58 represents a model of the auxiliary output port of a car. The output current is $i = 1 \text{ A}$ when $v = 0 \text{ V}$. The output voltage is $v = 12 \text{ V}$ when a $20\text{-}\Omega$ resistor is connected between the terminals. Suppose you wanted to charge a 12-V battery by connecting the battery at the port. How much current would the port deliver to the battery?

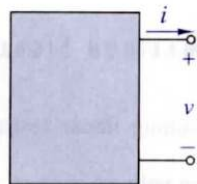


FIGURE P3-58

- 3-2 (a)** Formulate node-voltage equations for the circuit in Figure P3-2. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_B .
(c) Use these results to find v_x and i_x .

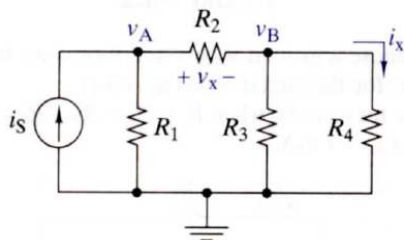


FIGURE P3-2

- 3-3 (a)** Formulate node-voltage equations for the circuit in Figure P3-3. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_B .
(c) Use these results to find v_x and i_x .

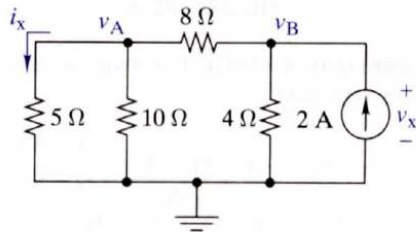


FIGURE P3-3

- 3-5 (a)** Formulate node-voltage equations for the circuit in Figure P3-5. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_C .
(c) Use these results to find v_x and i_x .

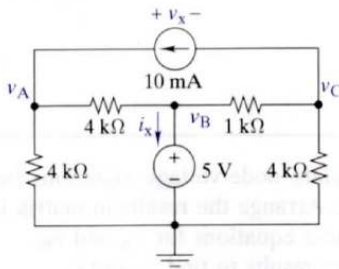


FIGURE P3-5

- 3-6 (a)** Choose a ground wisely and formulate node-voltage equations for the circuit in Figure P3-6.
(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$, $v_s = 25 \text{ V}$, and $i_s = 1 \text{ mA}$.

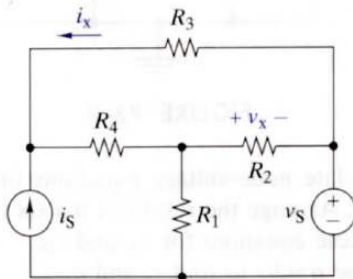


FIGURE P3-6

- 3-15(a) Formulate mesh-current equations for the circuit in Figure P3-15. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
 (b) Solve for i_A and i_B .
 (c) Use these results to find v_x and i_x .

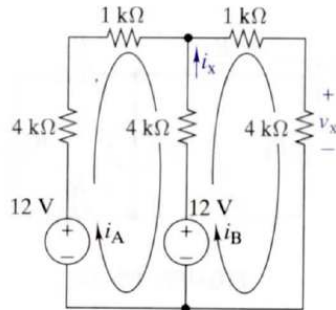


FIGURE P3-15

- 3-26(a) Formulate mesh-current equations for the circuit in Figure P3-26.
 (b) Formulate node-voltage equations for the circuit in Figure P3-26.
 (c) Which set of equations would be easier to solve? Why?
 (d) Use OrCAD to find the node voltages v_A and v_B and the mesh-currents i_A , i_B and i_C in Figure P3-26.

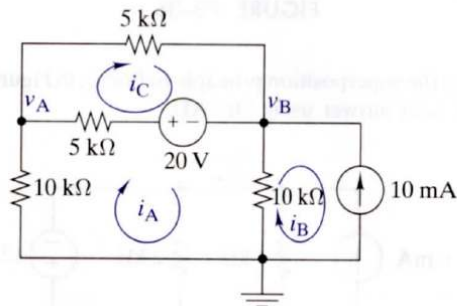


FIGURE P3-26

- 3-27(a) Formulate mesh-current equations for the circuit in Figure P3-27. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
 (b) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-27.
 (c) Formulate node-voltage equations for the circuit in Figure P3-27. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
 (d) Use MATLAB and node-voltage analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-27 and compare the effort required with each technique, i.e. mesh-current versus node-voltage.
 (e) Use OrCAD to verify your results in the previous two parts.

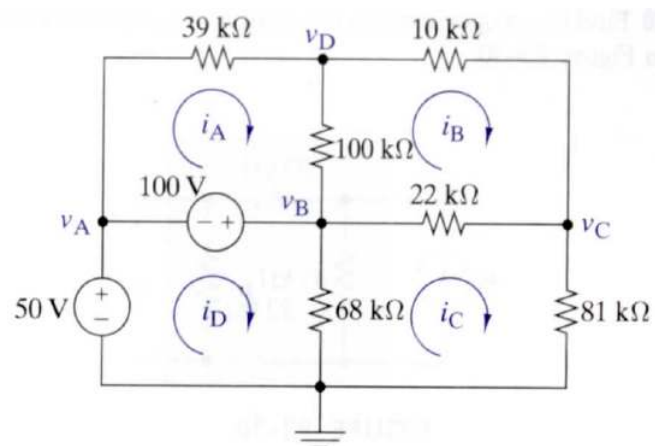


FIGURE P3-27